

24. If the stars are seen to rise perpendicular to the horizon by an observer, he is located on the –

- (a) Equator (b) Tropic of Cancer
- (c) South Pole (d) North Pole

I.A.S. (Pre) 2001

Ans. (a)

Stars move parallel to the celestial equator around North and South celestial poles. The celestial North and South poles are straight overhead to the geographical North and South Poles and celestial equator is almost at overhead to the geographical Equator and at 90° to the Poles. It creates occlusion from the points of East and West at the horizon. The degrees of celestial poles are equal in number to latitude of the equator for the horizon. If an observer sees the stars perpendicular to the horizon he is located at the equator.

25. A person stood alone in a desert on a dark night and wanted to reach his village which was situated 5 km East of the point where he was standing. He had no instrument to find the direction but he located the pole star. The most convenient way now to reach his village is to walk in the

- (a) Direction facing the pole star
- (b) Direction opposite to the pole star
- (c) Direction keeping the pole star to his left
- (d) Direction keeping the pole star to his right

I.A.S. (Pre) 2012

Ans. (c)

The Pole star shines in the north direction. Hence, in order to move towards the east, Pole star should always be on the left side.

26. The group of stars that indicates the direction of pole, is

- (a) Saptarishi (b) Mrig
- (c) Scorpio (d) Taurus

41st B.P.S.C. (Pre) 1996

Ans. (a)

Saptarishi is a group of stars that indicates the direction of the pole. Hence, option (a) is the correct answer.

27. Which one of the following is not related to space terminology?

- (a) Telemetering (b) Weightlessness
- (c) Cislunar (d) Byte

U.P.P.C.S. (Pre) 1997

Ans. (d)

A byte is a group of eight binary digits or bits operated on as a unit in computer terminology. The Rest of all are related to space terminology.

28. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of

- (a) Observation and understanding of the Universe
- (b) Study of the solar and the lunar eclipses
- (c) Placing satellites in the orbit of the Earth
- (d) Origin and evolution of living organisms on the Earth

I.A.S. (Pre) 2017

Ans. (a)

Event Horizon - It is a boundary that defines the area surrounding a black hole in space. Beyond this boundary events cannot affect an outside observer.

Singularity - It is a place in space-time where the gravitational field of celestial bodies becomes infinite.

String Theory - String theory states that all matter in the universe is composed of tiny one-dimensional strings, not point particles.

Standard Model - The Standard model of particle physics is the theory describing three of the four known fundamental forces (the electromagnetic, weak and strong) in the universe as well as classifying all known elementary particles.

29. Which one of the following is not a natural unit of time?

- (a) Tropical year (b) Lunar month
- (c) Standard time (d) Day

Uttarakhand P.C.S. (Pre) 2016

Ans. (c)

Standard time is the official standard time of a region or country.

ii. The Solar System

*The family of the 'Sun' is known as the solar system. Eight major planets, dwarf planets, satellites and countless minor planets, asteroids, meteors, and comets orbit the Sun to form the solar system. The solar system dates back to about 4.6 billion years.

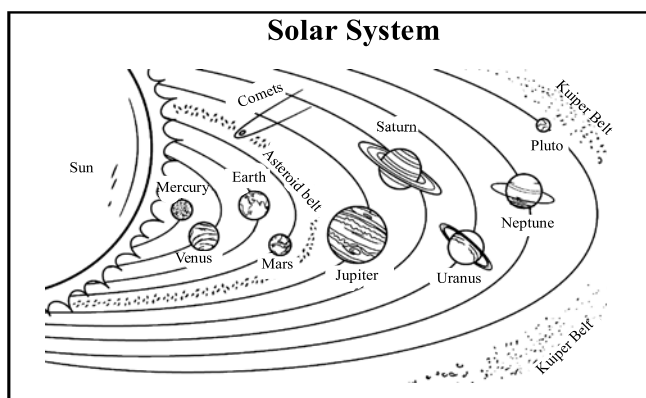
*Originally there were 9 planets in the solar system. Their average distance in ascending order from the Sun is as follows –

1. Mercury (57.9 million km)
2. Venus (108.2 million km)
3. Earth (149.6 million km)
4. Mars (228 million km)
5. Jupiter (778.5 million km)
6. Saturn (1432 million km)
7. Uranus (2867 million km)
8. Neptune (4515 million km)

9. Pluto (5906.4 million km)

Note- Between 14-25 August, 2006 the 26th General Assembly of the International Astronomical Union was held in Prague (Czech Republic). The assembly excluded Pluto from the list of planets and reclassified it as a dwarf planet.

*Planets differ from each other depending on their size, composition, temperature and time taken for a revolution. The main reason for this difference is their **distance** from the sun. *Some planets like Mercury, Venus, Earth and Mars are solid; these planets are called **terrestrial planets** whereas some planets have a gaseous composition like Jupiter, Saturn, Uranus and Neptune. Due to their large size these planets are also known as **Jovian Planets** (Jovian means like Jupiter). *Out of all the 8 planets, Earth has the maximum mean density i.e. 5.5 g/cm³. The mean density of other planets is as follows –



Mercury (5.4 g/cm³), Venus (5.2 g/cm³), Mars (3.9 g/cm³), Jupiter (1.3 g/cm³), Saturn (0.7 g/cm³), Uranus (1.3 g/cm³), Neptune (1.6 g/cm³).

*The Sun alone carries 99.8% of the total mass of the solar system. *Sun's mass is 1.989×10^{30} kilograms which is 3,33000 times that of the planet Earth. *Mercury is the **smallest planet** of the solar system whereas Jupiter is the **largest**. *Venus is the **brightest planet** whereas **Neptune** is the farthest. Neptune takes the maximum time in to complete one revolution around the Sun (165 yrs). *The descending order of **planets according to their size** is as follows - Jupiter, Saturn, Uranus, Neptune, Earth, Venus, Mars, Mercury. *The descending order of the **planets according to their mass** is as follows – Jupiter, Saturn, Neptune, Uranus, Earth, Venus, Mars & Mercury. *The ascending order of **planets according to their distance from the Sun**

is as follows – Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. *Among all the planets, Jupiter has the **maximum gravitational force** (23.1 m/s²). *Gases like hydrogen, helium, methane are found in abundance on planets like Jupiter and Saturn. The Goldilocks zone is also called the habitable zone. In astronomy, the Goldilocks zone is the range of the orbits around a star within which a planetary surface can support liquid water given sufficient atmospheric pressure. Liquid water is one of the most important elements in finding life on other planets. *At Present, Earth is the only planet in our solar system having a **Goldilocks zone**. This zone is neither too hot nor too cold. *In 1514 AD, Polish astronomer and mathematician **Nicolas Copernicus** presented a handwritten essay 'Commentariolus' (Little Commentary). In this essay, he gave information regarding the principle of revolution. According to this essay, the Sun is at the center of the universe and earth revolves around it. Later in 1543 AD, it was published by the name 'De revolutionibus orbium coelestium (On the Revolutions of the Heavenly Spheres)'. *It is noteworthy to mention that in 6th century A.D. i.e. 1000 years before Nicolaus Copernicus. The famous Indian astronomer **Varamihira** had already proposed the theory that, the Moon revolves around the Earth and Earth revolves around the Sun. ***Johannes Kepler** was a German astronomer, mathematician and astrologer. He is best known for propounding the 'three laws of planetary motion'. ***Galileo Galilei**, the famous scientist of Italy gave us the fundamental principles of falling bodies. *English scientist **Isaac Newton** gave us the fundamental principles of velocity and also discovered the laws of gravity.

1. Which of the following does not belong to the solar system?

- | | |
|---------------|------------|
| (a) Asteroids | (b) Comets |
| (c) Planets | (d) Nebula |

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The solar system was formed around 4.6 billion years ago. The solar system consists of the sun (star), cosmic dust or masses attached to each other by gravitational force, planets, dwarf planets, natural planetoids, asteroids, comets, meteoroids and cosmic dust while nebulae are not a part of the Solar System.

2. Who first propounded that the Sun is the center of our Solar system and the earth revolves around it?

- (a) Newton (b) Galileo
(c) Panini (d) Copernicus

M.P.P.C.S. (Pre) 1995

Ans. (d)

Nicolaus Copernicus was an astronomer and mathematician from Poland. In 1514, Copernicus propounded the Heliocentric Theory of the solar system in his essay 'Commentariolus' (Little Commentary). Notably, Indian astronomer Varahamihira propounded the same theory around a thousand years before Copernicus in the sixth century. He mentioned that the moon revolves around the earth and the earth revolves around the sun.

3. The scientist who first discovered that the earth revolves around the sun was –

- (a) Newton (b) Dalton
(c) Copernicus (d) Einstein

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

See the explanation of the above question.

4. Which scientist said for the first time that the earth revolves around the sun and that the earth is not the centre of the universe?

- (a) Galileo (b) Copernicus
(c) Isaac Newton (d) Kepler

M.P.P.C.S. (Pre) 1998

Ans. (b)

See the explanation of the above question.

5. Who first established that all the planets revolve round the sun?

- (a) Nicholas Copernicus (b) J. Kepler
(c) C. Galileo (d) D. Immanuel Kant

Jharkhand P.C.S. (mains) 2016

Ans. (a)

Nicholas Copernicus was the first to establish that all the planets including the Earth revolve around the sun.

6. Which one of the following statements is correct with reference to our solar system?

- (a) The Earth is the densest of all the planets in our solar system
(b) The predominant element in the composition of Earth is silicon

(c) The Sun contains 75 percent of the mass of the solar system

(d) The diameter of the Sun is 190 times that of the Earth

I.A.S. (Pre) 2002

Ans. (a)

The average density of Earth is highest among all the other eight planets of our Solar System. It is 5.5 gm/cm³. The density of other planets is as follows (Unit–gm/cm³) Mercury (5.4), Venus (5.2), Mars (3.9), Jupiter (1.3), Saturn (0.7), Uranus (1.3), Neptune (1.6). Thus statement (a) is correct. The largest constituent element of Earth is Iron (32.07%). Oxygen (30.12%) is the second and Silicon (15.12%) is the third largest element in the composition of Earth. The sun comprises 99.8% mass of the solar system. The diameter of the sun is about 109.2 times the Earth's diameter.

7. The number of the planet in the solar system is –

- (a) 7 (b) 9
(c) 12 (d) 21

U.P.P.C.S. (Pre) 1992

Ans. (b)

According to old recognition, there are a total of 9 planets which are in increasing order of distance from the sun, as follows :

- (i) Mercury (57.9 million km)
(ii) Venus (108.2 million km)
(iii) Earth (149.6 million km)
(iv) Mars (228 million km)
(v) Jupiter (778.5 million km)
(vi) Saturn (1432 million km)
(vii) Uranus (2867 million km)
(viii) Neptune (4515 million km)
(ix) Pluto (5906.4 million km)

Note : In the 26th General Assembly meeting of the International Astronomical Union held at Prague in 2006 an official definition of a planet was given and a resolution was passed by which Pluto has been removed from the group of 9 planets and now it is a dwarf planet. Hence, the total number of planets is 8 now.

8. Which of the given two planets are between Mars and Uranus in order of distance from the sun ?

- (a) Earth and Jupiter (b) Jupiter and Saturn
(c) Saturn and Earth (d) Saturn and Neptune

I.A.S. (Pre) 2008

Ans. (b)

See the explanation of the above question.

9. Arrange the planets given below in order of increasing distance from sun ?

- | | |
|------------|-----------|
| 1. Pluto | 2. Earth |
| 3. Jupiter | 4. Uranus |

Codes :

- | | |
|-------------|-------------|
| (a) 2,3,4,1 | (b) 4,3,2,1 |
| (c) 3,2,4,1 | (d) 1,2,4,3 |

U.P. Lower Sub. (Spl.) (Pre) 2003

U.P. Lower Sub. (Pre) 2002

Ans. (a)

See the explanation of the above question.

10. The Planets between Earth and Sun are –

- | | |
|-----------------------|------------------------|
| (a) Mars and Mercury | (b) Mars and Venus |
| (c) Mercury and Venus | (d) Jupiter and Saturn |

U.P.P.S.C. (GIC) 2010

Ans. (c)

See the explanation of the above question.

11. The Earth is located between –

- | | |
|-----------------------|-----------------------|
| (a) Venus and Mars | (b) Mars and Jupiter |
| (c) Venus and Jupiter | (d) Mercury and Venus |

U.P.P.C.S. (Mains) 2012

Ans. (a)

See the explanation of the above question.

12. Which is true about planets ?

- (a) Planets are non-luminous bodies and don't shine
 (b) Planets shine though they are non-luminous bodies
 (c) Planets do not shine though they are the luminous body
 (d) Planets are the luminous body and also shine

U.P. P.C.S. (Pre) 1992

Ans. (b)

Planets are non-luminous bodies but they shine because they reflect the light of the Sun. Thus option (b) is correct.

13. Match List-I with List-II and select the correct answer by using the codes given below the lists:

List-I (Special characteristic)

- A. Smallest planet of the solar system
 B. Largest planet of the solar system
 C. Planet second to the sun in the solar system
 D. Planet nearest to the sun

List- II (Name of the planet)

- | | |
|------------|----------|
| 1. Mercury | 2. Venus |
| 3. Jupiter | 4. Pluto |
| 5. Saturn | |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 3 | 5 | 1 |
| (b) | 3 | 5 | 1 | 2 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 4 | 3 | 2 | 1 |

I.A.S. (Pre) 1998

Ans. (d)

The correctly matched list is as follows :

(Special Characteristics) (Name of Planets)

Smallest planet of the - Pluto (After it was removed from Solar the group of planets Mercury is the smallest).

Largest planet of the solar - Jupiter system

Planet second to the Sun - Venus in the solar system

Planet nearest to the Sun - Mercury

14. The planet nearest to the Sun is –

- | | |
|-------------|-----------|
| (a) Mercury | (b) Earth |
| (c) Venus | (d) Pluto |

U.P.P.C.S. (Mains) 2004

Ans. (a)

See the explanation of the above question.

15. Which of the following pairs is not properly matched ?

- | | |
|---|-----------|
| (a) The largest planet of the Solar System | - Jupiter |
| (b) The smallest planet of the Solar System | - Mercury |
| (c) The brightest planet of the Solar System | - Venus |
| (d) The slowest moving planet of Solar System | - Mars |

U.P.P.C.S. (Mains) 2011

Ans. (d)

All of the given options are true, except option (d) because the slowest moving planet of the solar system is Neptune. Its orbital velocity is lowest among all planets and takes 60190 Earth days (approx 165 years) to complete one revolution around the sun.

16. Arrange the following in decreasing order of size and select the correct answer from the codes given below :

- | | |
|------------|-----------|
| 1. Jupiter | 2. Uranus |
| 3. Earth | 4. Saturn |

Code :

(a) 1, 4, 3, 2

(b) 4, 1, 2, 3

(c) 1, 4, 2, 3

(d) 4, 1, 3, 2

M.P.P.C.S. (Pre) 2008

Ans. (c)

The descending order of the size of the given planets is as follows—

Jupiter > Saturn > Uranus > Earth. Size of planets in decreasing order is as follows. Jupiter, Saturn, Uranus, Neptune, Earth, Venus, Mars, Mercury.

17. Hydrogen, Helium and Methane are the main gases present on

- (a) Uranus, Neptune and Pluto
- (b) Jupiter, Saturn and Mars
- (c) Uranus, Neptune and Venus
- (d) Mars and Venus

U.P.P.C.S. (Mains) 2004

Ans. (*)

Hydrogen, Helium and Methane gases are present on Jupiter, Saturn, Uranus and Neptune.

18. Which of the following is not a planet of Solar System?

- (a) Mercury
- (b) Florida
- (c) Venus
- (d) Saturn

M.P.P.C.S. (Pre) 2010

Ans. (b)

Mercury, Venus and Saturn are the planets of the solar system while Florida is a state of the United States of America.

19. 'The term 'Goldilocks Zone' is often seen in the news in the context of

- (a) The limits of the habitable zone above the surface of the Earth
- (b) Regions inside the Earth where shale gas is available
- (c) Search for the Earth-like planets in outer space
- (d) Search for meteorites containing precious metals

I.A.S. (Pre) 2015

Ans. (c)

In astronomy, the circumstellar habitable zone is also called 'Goldilocks Zone'. It refers to the habitable zone around a star where the temperature is just right- not too hot and not too cold for liquid water to exist on a planet. Earth is the only planet inhabiting life in the entire solar system. It is the only planet situated in the Goldilocks zone of the solar system.

20. Match list – I with list –II and choose the correct answer by using codes given below –

List – I

(A) Planet

(B) Satellite

(C) Comet

(D) Planetoid or artificial Satellite

Code :

	A	B	C	D
(a)	2	1	4	3
(b)	1	2	3	4
(c)	4	3	1	2
(d)	2	1	3	4

List- II

1. Moon

2. Uranus

3. Mariner

4. Halley

U.P.P.C.S. (Pre) 1999

Ans. (a)

Uranus is a planet, Moon is a satellite (natural), Halley is a comet and Mariner is an artificial satellite.

21. Which of the following statements are correct regarding the solar system?

I. Mercury is the hottest planet in the solar system

II. Ganymede, satellite of Saturn, is the largest satellite in the solar system

III. Neptune is surrounded by methane gas rings of sub-zero temperature

IV. Phobos and Deimos are two satellites of Mars

- (a) Only I and II are correct
- (b) Only II and III are correct
- (c) Only III and IV are correct
- (d) I, II, III and IV all are correct

Chhattisgarh P.C.S. (Pre) 2020

Ans. (c)

Venus is the hottest planet in the solar system, not Mercury. This is, because its dense atmosphere acts as a greenhouse and heats the surface to about 464°C.

Ganymede is the largest satellite in the solar system, but it is a satellite of Jupiter, not Saturn.

Neptune is surrounded by methane gas rings of sub-zero temperature.

Mars has two small moons : Phobos and Deimos.

22. Which is the chief heavenly body of solar system?

- (a) Earth
- (b) Jupiter